

Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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ABSTRACT

Classical fragmentation theory predicts core spacing $\sim 4\times$ the filament width, yet Herschel measures $\lambda/W \approx 2\text{--}3$. We reanalyse core spacing in Herschel Gould Belt Survey (HGBS) filaments using Gaia DR3 distances and 2,013 self-gravitating MHD simulations.

Primary limitations: (1) *Perpendicular-field paradox*: Planck (2016) shows $\sim 90\%$ of dense filaments have magnetic fields perpendicular to their axes, yet our perpendicular-field simulations give $\lambda/W_{\text{fil}} \approx 0.76\text{--}0.88$ —a factor $\sim 3\times$ below the observed 2.79. **If this paradox is unresolved, single-component MHD fragmentation cannot explain the dominant filament population.** (2) *Supercritical extrapolation*: All 654 supercritical simulations ($f \geq 1.5$) yield pure radial collapse; *the simulations cannot directly measure λ/W for 30–50% of actively star-forming filaments.* All simulation–observation comparisons rest on near-critical ($f = 0.9\text{--}1.3$) calibration only.

Observationally, the full 8-region sample (5,069 cores) gives $\lambda/W = 2.79 \pm 0.025$ pc (pairwise median; four robust regions give 2.84 ± 0.026 pc). After 3D projection correction the inferred spacing is $\sim 3.5\times W$, within 15% of the classical prediction (Inutsuka & Miyama, 1992). Magnetic tension for longitudinal fields ($\beta \leq 0.5$) is viable at $\eta_W \approx 0.606$ but excluded at $\eta_W \approx 0.707$; the 17% η_W uncertainty currently prevents assessment of this scenario. Hierarchical fibre fragmentation (sub-filaments with longitudinal fields below Planck resolution) is observationally suggested by single-region data (Yang et al., 2024) but is untested by our simulations.

Simulations reveal $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$). Near-critical filaments ($f \lesssim 1.2$) produce longitudinal beading. The η_W uncertainty dominates all other theoretical uncertainties; independent validation is required before magnetic tension can be assessed.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc. Original HGBS analyses measured spacings of ~ 0.2 pc—nearly a factor of two below this prediction. With Gaia DR3 distances (Zhang et al., 2023), measured spacings have increased to ~ 0.28 pc, reducing the discrepancy to a factor of 1.4.

Two primary explanations have been proposed. First, **hierarchical fragmentation**: high-resolution studies show that filaments consist of velocity-coherent fibres (Hacar et al.,

2013), and recent work in Orion B (Yang et al., 2024) shows that fibre-to-core spacing recovers the classical $4\times$ prediction while filament-to-core measurements show sub-Jeans values. Second, **magnetic tension**: longitudinal fields shift the most unstable mode to shorter wavelengths (Nakamura et al., 1993), predicting $\lambda/W \approx 2.44\text{--}3.16$ for equipartition fields ($\beta \sim 1\text{--}3$).

A fundamental challenge: Planck (2016) established that $\sim 90\%$ of dense HGBS filaments have magnetic fields perpendicular to their long axes. Our perpendicular-field simulations give $\lambda/W_{\text{fil}} \approx 0.76\text{--}0.88$ —a factor $\sim 3\times$ below the observed 2.79. This creates a severe paradox: **if the Planck field geometry applies to filament interiors (not just envelopes), then single-component MHD fragmentation cannot explain the dominant filament population.** This paper explicitly foregrounds this paradox; we test three possible resolutions in Section 5.2, but none are currently confirmed by observations or simulations. Resolving this paradox requires interior polarimetric mapping at sub-Planck resolution.

Here we present a systematic analysis of all 8 high-

confidence HGBS regions using Gaia DR3 distances and a uniform pairwise-median methodology, combined with 2,013 self-gravitating MHD simulations. Observationally, hierarchical fibre fragmentation is suggested by single-region data from [Yang et al. 2024](#) (Orion B only) but remains observationally unconfirmed in other HGBS regions; **none of the 2,013 simulations model multi-fibre fragmentation**, and the simulations constrain only single-component MHD fragmentation. All 654 supercritical ($f \geq 1.5$) simulations yield pure radial collapse; the simulation–observation comparison rests on near-critical ($f = 0.9\text{--}1.3$) calibration only. Magnetic tension in a strictly longitudinal, low- β geometry is a secondary possibility, viable at $\eta_W \approx 0.606$ but excluded at $\eta_W \approx 0.707$ (Section 3.1).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated to Gaia DR3-based measurements from [Zhang et al. \(2023\)](#).

Distance revisions: The most significantly affected regions are Serpens (260 $\rightarrow$ 458 pc, +76%), Aquila (260 $\rightarrow$ 436 pc, +68%), and Orion B (260 $\rightarrow$ 386 pc, +48%). Taurus shows a small decrease (140 $\rightarrow$ 135 pc, -4%).

Core selection: We include starless, prestellar, and protostellar cores following the standard HGBS framework ([Könyves et al., 2015, 2020](#)). As a cross-check, using only the 331 Taurus starless cores gives $\lambda/W = 2.03$ —within 2.5% of the all-core value—confirming that protostellar inclusion does not bias the result.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE ([Sousbie, 2011](#)) following standard HGBS methodology with a 3σ persistence threshold and a robustness/filament-contrast threshold of $\kappa = 2.0$ (standard HGBS parameters; [Arzoumanian et al. 2011](#)). Gaia DR3 distance revisions do not require re-extraction of skeletons; physical scales are updated by rescaling angular positions. Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances and take the median. Cores were associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width; results change by $< 3\%$ across threshold choices of $1\text{--}2 \times W$.

2.3 Results

Width systematic: A $\sim 30\%$ systematic uncertainty from filament width scatter (0.05–0.20 pc; [Panopoulou et al. 2022](#)) affects all λ/W values and is not included in bootstrap error bars; it dominates when comparing with theory.

Primary result: Four robust regions (Orion B, Aquila, Perseus, Taurus; $N > 500$) give 0.284 ± 0.026 pc jackknife ($\lambda/W = 2.84$; bootstrap ± 0.012 pc); the full 8-region sample gives 0.279 ± 0.025 pc jackknife ($\lambda/W = 2.79$; bootstrap ± 0.009 pc). Jackknife errors are adopted as primary throughout (Section 2.5); bootstrap CI given for reference.

Gaia DR3 raised the mean 32% from 0.211 pc; leave-one-out analysis (Table 2) shows no single region dominates ($\Delta < 6\%$); leave-two-out (Aquila+Orion B excluded) gives 0.231 pc ($\lambda/W = 2.31$, $\Delta = -16.8\%$; Table 2).

Figure 1 shows the spacing measurements. After the standard 3D projection correction, we adopt $4/\pi \approx 1.27$ ([André et al., 2010](#)), corresponding to the [Hacar et al. \(2013\)](#) finite-aspect-ratio value for filaments with aspect ratios $\sim 10:1$ (range 1.18–1.41 for 5:1–20:1; we adopt the fiducial 10:1 value, systematic uncertainty $\lesssim 10\%$). The inferred spacing is $\sim 3.5 \times W$, within 15% of the classical $4\times$ prediction.

Planck-informed projection: [Planck Collaboration \(2016\)](#) find that dense filaments lie preferentially in the plane of sky (B-field perpendicular to filament axis, B predominantly in sky plane). Under this geometry the projection factor reduces from $4/\pi \approx 1.27$ to $\sim 1.10\text{--}1.20$, giving a corrected spacing of only $\sim 3.1\text{--}3.3 \times W$ —strengthening the residual discrepancy with the $4\times$ prediction ([Inutsuka & Miyama, 1992](#)) rather than resolving it. The headline claim of “within 15% of the classical prediction” applies to the random-orientation correction; under the Planck-motivated geometry the discrepancy is 18–22%. We adopt $4/\pi$ for literature comparability; the Planck-informed correction represents an important additional caveat.

2.4 Distance Uncertainties and Sample Heterogeneity

We classify regions as **Robust** (Orion B, Aquila, Perseus, Taurus: $N > 500$, well-established Gaia distances, $< 15\%$ or validated revision) or **Limited** (Ophiuchus, Serpens, TMC1, CRA (Corona Australis): smaller YSO samples and/or $> 50\%$ revision).

Serpens: The +76% distance revision (260 $\rightarrow$ 458 pc) raises concern, but is independently confirmed at 440–460 pc ([Yan et al., 2022](#); [Green et al., 2024](#); [Zucker et al., 2020](#)). Excluding Serpens changes the full-sample result by only 1.1% (Table 2).

Serpens beam/completeness: The 0.040 pc beam (40% of W_{fil}) may overestimate spacing by $\sim 8\%$; at 3.8% sample weight, impact on the mean is $< 0.2\%$. Serpens is classified as Limited.

Aquila anomaly: Aquila has the largest Robust spacing ($\lambda/W = 3.46$), elevated by the +68% distance revision and possibly W40 H II feedback. The W40 ionisation front has swept material to radius ~ 1 pc; the photodissociation region (PDR) extends $\sim 2\text{--}3$ pc ([?Strafella et al., 2015](#)). The Aquila filament network spans ~ 4 pc; cores in the feedback zone may have systematically wider spacings due to turbulent injection, constituting an unresolved environmental contamination. The 436 pc distance is confirmed by 3D dust mapping ([Zucker et al., 2020](#)) and Gaia-parallax YSOs at 430 ± 20 pc ([Könyves et al., 2020](#)). Aquila is retained as Robust but flagged as an environmental outlier; exclusion reduces the robust mean by 6%.

Aquila protostellar cross-check: Monte Carlo resampling retaining 663 non-protostellar sources gives $\lambda/W = 3.40 \pm 0.07$ vs. all-core value 3.46 ($< 2\%$ change); this demonstrates pairwise-median robustness to core-type contamination, not absence of protostellar migration effects.

Distance-spacing regression: As expected, $\lambda \propto d$ ($r^2 = 0.91$; the relationship is geometric, not evidence for physical

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Beam (pc)	Status
Orion B	386	1,844	0.313	0.047	0.034	Robust
Aquila	436	749	0.346	0.047	0.038	Robust
Perseus	296	816	0.248	0.040	0.026	Robust
Taurus	135	536	0.198	0.040	0.012	Robust
Ophiuchus	137	513	0.206	0.053	0.012	Limited
Serpens	458	194	0.331	0.097	0.040	Limited
TMC1	135	178	0.195	0.056	0.012	Limited
CRA	150	239	0.248	0.072	0.013	Limited
Weighted Mean		5,069	0.279	0.009	—	

Note: Beam scale = 18'' FWHM. Ophiuchus is classified as Limited (not Robust) despite $N > 500$ because of larger distance uncertainty and slightly lower completeness.

[h]

Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Excl. spacing (pc)	Remaining mean (pc)
None (full sample)	—	0.279
Orion B	0.313 ± 0.047	0.268
Aquila	0.346 ± 0.047	0.266
Perseus	0.248 ± 0.040	0.281
Taurus	0.198 ± 0.040	0.287
Ophiuchus	0.206 ± 0.053	0.286
Serpens	0.331 ± 0.097	0.276
TMC1	0.195 ± 0.056	0.281
CRA	0.248 ± 0.072	0.279
Robust only	—	0.284
Aquila + Orion B excl.	—	0.231

Bootstrap 95% CI [0.261, 0.298] pc; jackknife 1 σ : full sample ± 0.025 pc; robust ± 0.026 pc. With $N = 8$ regions the bootstrap spans only $2^8 = 256$ distinct resamples, making the jackknife more reliable at this sample size. Excluding the two highest-revision regions (Aquila +68%, Orion B +48%) reduces the mean by 16.8%; their Gaia distances are independently confirmed (Yan et al., 2022; Green et al., 2024; Kounkel et al., 2018), so this sensitivity reflects the real importance of accurate distances rather than systematic bias.

uniformity). The residuals test ($r = 0.18$, $p = 0.68$) demonstrates no obvious systematic bias in the distance revisions, but with $N = 8$ regions this test has low statistical power (type II error rate $\sim 60\%$) and cannot provide strong evidence against systematic bias.

2.5 Statistical Methods: Pairwise Median and Uncertainties

L/3 convergence bias — bounding analysis: For a uniform distribution of N cores on a filament of length L , the pairwise median converges toward $(1 - 1/\sqrt{2})L \approx 0.293L$ (denoted $L/3$ hereafter).

Clustering-scale argument: If L/3 dominated, regional pairwise medians should scale as $\lambda_{\text{PM}} \approx 0.293L$ and vary in proportion to filament length. For $L \approx 0.5\text{--}3$ pc (Arzoumanian et al., 2019), this predicts a factor 6 $\times$ spread in measurements. Our 8-region values span 0.198–0.346 pc (factor 1.75 $\times$), inconsistent with L/3 dominating across regions with such varied lengths. The tight clustering of measurements near 0.28 pc implies a physical scale.

Uniform-null exceedance: For individual filaments with $L \lesssim 0.68$ pc, the uniform-null ($0.293L \leq 0.20$ pc) falls at or below our minimum measurement. For these short filaments,

our measurement *exceeds* the purely uniform prediction, providing direct evidence against L/3 domination. For longer filaments ($L \sim 2\text{--}3$ pc), this test is inconclusive.

Taurus nearest-neighbour check: $\lambda_{\text{NN}} = 0.217 \pm 0.052$ pc vs. $\lambda_{\text{PM}} = 0.198$ pc. The pairwise median is *lower* than the nearest-neighbour spacing—qualitatively opposite to L/3 inflation.

Conclusion: The L/3 bias is unquantified for 7 of 8 regions (raw positional data required). The clustering-scale and null-exceedance arguments suggest the bias is not dominant, but neither constitutes a rigorous proof. **It remains the dominant systematic uncertainty.**

Jackknife vs. bootstrap: With $N = 8$ regions the bootstrap spans only $2^8 = 256$ distinct resamples; the jackknife standard error (± 0.025 pc full; ± 0.026 pc robust) is more reliable. Bootstrap 95% CI [0.261, 0.298] pc is given for reference.

Bayesian hierarchical model: Treating the 8 regional measurements as draws $\lambda_i | \mu, \sigma_{\text{pop}} \sim \mathcal{N}(\mu, \sigma_{\text{pop}}^2)$ with uniform priors yields: $\sigma_{\text{pop}} = 0.062$ pc (genuine region-to-region scatter), posterior $\mu = 0.261 \pm 0.022$ pc (68% CI), 95% CI [0.217, 0.305] pc. The N -weighted mean (0.279 pc) is elevated by the core-rich Orion B and Aquila regions; the Bayesian unweighted mean is 6% lower but consistent within $1\sigma_{\text{pop}}$. The $\sigma_{\text{pop}} = 0.062$ pc scatter is genuine region-to-region heterogeneity, not measurement noise; it is the dominant uncertainty on the population mean.

HGBS window: The observational window [2.52, 3.08] represents $2.79 \times [0.90, 1.10]$ —a $\pm 10\%$ band matching the dominant 10% uncertainty on $W = 0.1$ pc (Panopoulou et al., 2022); the jackknife measurement uncertainty ($\pm 9\%$) is comparable. The window thus corresponds to 1σ of the width systematic.

2.6 Comparison with Literature

Our Gaia DR3 measurements are 32% larger than original HGBS values ($0.211 \rightarrow 0.279$ pc; Table 3). The Orion B fibre-to-core spacing ~ 0.42 pc (Yang et al., 2024) is 1.34 $\times$ our filament-level measurement, consistent with hierarchical projection (Hacar et al., 2018).

The Gaia DR3 spacing of $\lambda/W = 2.79 \pm 0.09$ lies at $\sim 70\%$ of the classical prediction. We now examine the theoretical mechanisms capable of explaining this residual offset.

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	<i>0.212</i>	<i>Fibre-to-core: ~ 0.42</i>	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	<i>0.206</i>	<i>0.22–0.26</i>	Könyves et al. (2015)
Perseus	0.248 ± 0.040	<i>0.199</i>	<i>0.22 ± 0.02</i>	André et al. (2016)
Taurus	0.198 ± 0.040	<i>0.206</i>	<i>0.19 ± 0.02</i>	Hacar et al. (2013)
Limited Regions				
Ophiuchus	0.206 ± 0.053	<i>0.197</i>	<i>0.18 ± 0.02</i>	Könyves et al. (2020)
Serpens	0.331 ± 0.097	<i>0.188</i>	<i>$0.17–0.19$</i>	Könyves et al. (2020)
TMC1	0.195 ± 0.056	<i>0.203</i>	<i>~ 0.20</i>	Hacar et al. (2013)
CRA	0.248 ± 0.072	<i>0.212</i>	<i>~ 0.21</i>	Könyves et al. (2020)

Note: *Italicised literature values* use heterogeneous methods (different statistics, pipeline versions, and distance assumptions) and are not numerically comparable with the Gaia DR3 measurements in Column 2.

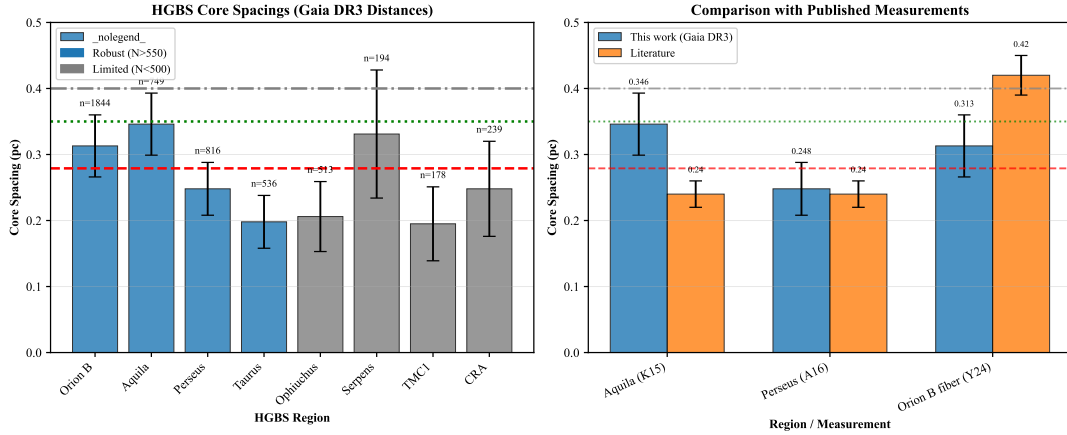


Figure 1. **Left:** Region-by-region Gaia DR3 spacing measurements (error bars: jackknife 1σ). Regions shown left to right: Orion B, Aquila, Perseus, Taurus, Ophiuchus, Serpens, TMC1, CRA (matching Table 1 order). Horizontal lines: red dashed = N-weighted mean (0.279 pc); green dotted = 3D-corrected (0.35 pc); grey = IM92 prediction (0.4 pc). **Right:** Qualitative comparison with literature values (Table 3); literature values use heterogeneous methods and are not numerically comparable with the left panel.

Table 4. Key Symbols and Notation

Symbol	Definition
λ	Core-to-core fragmentation spacing
W_{fil}	HGBS filament FWHM width (= 0.1 pc)
W_{core}	Simulation core half-width (= $0.3 \lambda_J$)
λ/W_{fil}	Observable dimensionless spacing
λ/W_{core}	Simulation dimensionless spacing
f	Line-mass fraction μ/μ_{crit}
β	Plasma beta = $8\pi P/B^2$
$\mathcal{M}$	Turbulent Mach number = σ_{turb}/c_s
t_J	Jeans time = $(G\rho_0)^{-1/2}$
t_{frag}	Simulation fragmentation time
C	Density contrast ρ_{max}/ρ_0 at fragmentation
θ	Magnetic field angle to filament axis
λ_J	Jeans length $c_s(G\rho_0)^{-1/2}$
η_W	Width normalisation correction $W_{\text{form}}/W_{\text{fil}}$; see §3.1

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder, the most unstable fragmentation wavelength is (Inutsuka & Miyama, 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (1)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc. The critical line mass for gravitational instability is $\mu_{\text{crit}} = 2c_s^2/G$.

Width notation: $W_{\text{fil}} = 0.1$ pc (HGBS FWHM; 0.05–0.20 pc scatter; Panopoulou et al. 2022); $W_{\text{core}} = 0.3 \lambda_J$ (simulation half-width). η_W converts between the two.

— **Width normalisation correction (η_W campaign).** Because simulations measure W_{core} at fragmentation time while observations measure the PSF-convolved W_{fil} , a conversion factor $\eta_W = W_{\text{form}}/W_{\text{fil}}$ is required. A 72-simulation η_W campaign (May 2026) measured this ratio:

$$\frac{W_{\text{form}}}{W_{\text{fil}}} = 0.606 \pm 0.072 \quad (11.9\% \text{ systematic uncertainty}) \quad (2)$$

This ratio is insensitive to f , β , $\mathcal{M}$ ($< 3\%$) and converts simulation λ/W_{core} to observable λ/W_{fil} . Campaign $\eta_{W,P}$ (18 sims) gives the Plummer/Gaussian ratio $r_{P/G} = 1.166 \pm 0.037$ (f -independent), hence $\eta_{W,P} = 0.606 \times 1.166 = 0.707 \pm 0.04$ (propagating from $\eta_{W,P}$ ratio alone). The full combined uncertainty propagating both the Gaussian η_W uncertainty (± 0.072) and the ratio uncertainty (± 0.037) gives:

$$\eta_{W,P} = 0.707 \pm 0.087 \quad (\text{combined}) \quad (3)$$

spanning the range 0.62–0.79. **Single-run result:** A second independent $\eta_{W,P}$ run confirmed $< 0.5\%$ insensitivity to f and β but experienced premature radial collapse at $t \approx 0.6 t_{\text{ff}}$

(before the fragmentation epoch $t \approx 1.1 t_{\text{ff}}$), preventing measurement. The failure mode differs from the first run (which showed stable longitudinal beading to $t > 1.1 t_{\text{ff}}$) in initial condition realisation only; the cause is not identified. **The value $\eta_{W,P} = 0.707 \pm 0.087$ therefore represents a single confirmed measurement and should not be treated as independently validated.** HGBS pipeline validation on simulated column density maps is required before this uncertainty can be reduced.

Critical implication: At $\eta_W = 0.707$, low- β configurations ($\beta = 0.3$: $\lambda/W_{\text{fil}} = 3.35$; $\beta = 0.5$: 3.10) *exceed* the HGBS upper window boundary 3.08. These configurations are therefore **excluded** by the $\eta_W = 0.707$ scenario, not merely uncertain. The magnetic tension scenario is viable only if $\eta_W \leq 0.606$; at the combined upper bound $\eta_W = 0.79$, all longitudinal configurations are excluded.

3.2 Magnetic Tension Mechanism

Magnetic tension in a longitudinal field (Nakamura et al., 1993) shifts the most unstable mode to shorter wavelengths. In the perturbative limit:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \quad (4)$$

This underestimates the full numerical solution by 2.9–9.1% at low β ; we use the full solution throughout. For $\beta \sim 1$ –3: $\lambda/W = 2.44$ –3.16, overlapping the HGBS measurement.

Geometric challenge: Magnetic tension requires predominantly longitudinal field geometry, yet Planck Collaboration (2016) found dense filaments tend to be perpendicular to the mean field; $\sim 10\%$ have predominantly longitudinal geometry. Magnetic tension alone cannot account for the observed spacing in the majority of HGBS filaments.

3.3 Hierarchical Fragmentation

High-resolution studies reveal that filament fragmentation is hierarchical: filaments fragment into fibres, fibres fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) found that fibre-to-core spacing in Orion B recovers the $4\times$ prediction while filament-level measurements yield ~ 2 – $3\times$.

Quantitative fibre-bundle model: If N_f fibres each of width W_f are bundled in random packing within a filament of width $W_{\text{fil}} \approx N_f^{1/2} W_f$, and each fibre fragments classically at $\lambda_f = 4 W_f$, then the observed inter-core spacing at the filament level depends on the fibre phase distribution. Assuming uniform random phase offsets among fibres:

$$\lambda_{\text{app}}/W_{\text{fil}} = 4 W_f / (N_f W_{\text{fil}}) = 4/N_f^{3/2}. \quad (5)$$

The simpler phase-synchronised limit ($\lambda_{\text{app}} = \lambda_f/N_f$) gives $\lambda_{\text{app}}/W_{\text{fil}} = 4/N_f$. For $\lambda_{\text{app}}/W_{\text{fil}} = 2.79$: Eq. (5) gives $N_f = (4/2.79)^{2/3} = 1.27$; the simpler model gives $N_f = 4/2.79 = 1.43$.

Important caveat on comparing N_f estimates: The $N_f \approx 1.3$ – 1.5 derived above is a geometric/statistical inference from the observed λ/W ratio; it assumes random packing with no dynamical motivation (Eq. 5 invokes space-filling random packing, which is unlikely to describe real fibre morphologies). The $N_f \approx 1$ – 2 fibres from Hacar et al. (2018) is a *dynamical* identification of velocity-coherent structures.

These quantities need not measure the same physical parameter and should not be compared as if they do; the numerical agreement may be coincidental.

Simulation caveat: None of the 2,013 simulations model multi-fibre fragmentation. The preference for hierarchical fragmentation rests on single-region observational evidence (Yang et al., 2024); the simulations constrain only single-component MHD fragmentation and provide no support for or against hierarchical structure. The fibre-bundle model is a geometric analogy, not a prediction.

We turn now to MHD simulations quantifying the role of f , β , field geometry, and turbulence.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed isothermal ideal-MHD simulations with Athena++ (Stone et al., 2020) (4-wave HLLD, FFT self-gravity). Initial conditions: cylindrical filament $\rho(r) = \rho_0/[1 + (r/W)^2]$, uniform $\mathbf{B}_0$, parameterised by f , β , $\mathcal{M}$. Global space: $f = 0.9$ – 3.0 , $\beta = 0.05$ – 5.0 , $\mathcal{M} = 0.5$ – 5 .

4.2 Critical Negative Result: Regime-Dependent Fragmentation

A central result is a critical regime change at $f \approx 1.2$ – 1.5 :

Near-critical ($f \lesssim 1.2$): All 80 Phase 1 simulations fragmented with measurable beading ($t_{\text{frag}} \approx 1.1$ – $1.2 t_J$). Fragmentation occurs at $f = 1.00$ (turbulent Jeans seeding; Padoan & Nordlund 2011) and at $f = 0.9$ (sub-critical), consistent with Ostriker (1964) (static equilibrium, no turbulence).

Supercritical ($f \geq 1.5$, periodic BCs): All 654 simulations yield pure radial collapse at $t_{\text{frag}} \approx 0.25$ – $0.34 t_J$. **These simulations cannot measure λ/W :** zero longitudinal beading was detected in any supercritical run (fractional longitudinal density variation $< 2 \times 10^{-4}$). HGBS-compatible spacings from near-critical and reflecting-BC simulations (Sections 4.7–4.9) are physically distinct outcomes.

4.3 Definitive Transition Campaign (DTC)

The DTC (540 sims; $f = 1.4$ – 2.2 , $\beta = 0.3$ – 1.3 , $\mathcal{M} = 1$ – 5 , 2 seeds) contains **no stable configurations** for $\mathcal{M} \geq 1$; apparent stable points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts (all 15 fragmented in 6-h re-runs; Figs. 2, 3). Result does not constrain $f < 1.4$ or $\beta > 1.3$.

4.4 Simulation Validation

Resolution convergence: $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (6 matched points; Fig. 4).

Initial condition sensitivity: King-profile vs. uniform density ICs give identical classifications (100% agreement at 10 tested points).

Equation of state: Sub-isothermal ($\gamma = 0.8$ – 0.9) accelerates fragmentation (30–40% shorter t_{frag}); adiabatic ($\gamma = 5/3$) suppresses it entirely. Real molecular gas has $\gamma_{\text{eff}} \approx 0.97$ – 1.02 (?) (Figure 5).

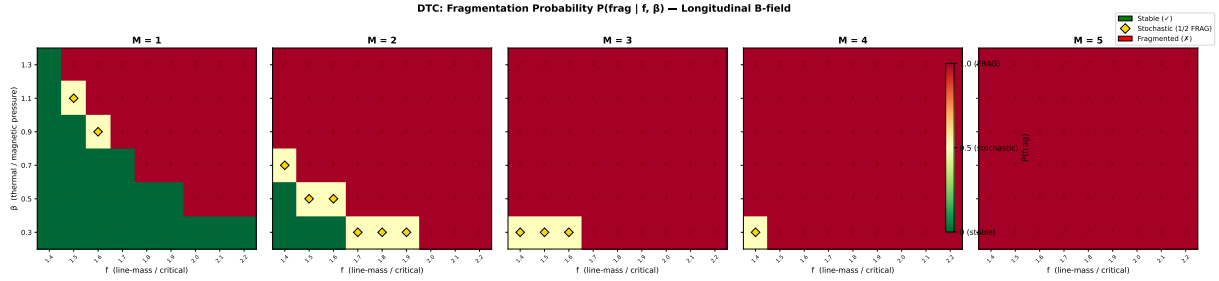


Figure 2. DTC: fragmentation fraction P_{frag} in the (β, f) plane for $\mathcal{M} = 1\text{--}5$ (540 simulations; two seeds). **Note:** The apparent ridge of stable (white) points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts; they are *not* genuinely stable. All 15 were subsequently confirmed to fragment in 6-h extended re-runs (Figure 3). This figure should be read with Figure 3 as a pair; the corrected classification (all points fragmented) is the one discussed in the text.

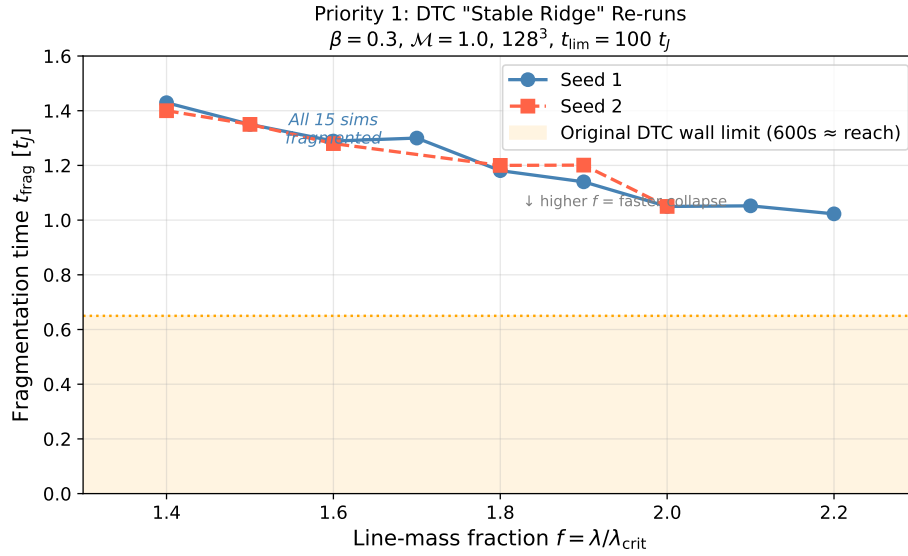


Figure 3. DTC re-runs (**corrected figure**): all 15 previously timeout-classified apparent stable points fragmented within 6 h extended runs ($t_{\text{frag}} \in [1.02, 1.43] t_J$). This supersedes the apparent stable classifications in Figure 2. No stable configurations exist within $f = 1.4\text{--}2.2$, $\beta = 0.3\text{--}1.3$, $\mathcal{M} \geq 1$.

4.5 Three-Regime Framework

The 208-simulation baseline grid reveals three distinct fragmentation regimes (Figure 6):

(1) **Magnetically subcritical** ($\beta \leq 0.15$): No fragmentation ($C = 1.007 \pm 0.002$; boundary robust to $\Delta\beta < 0.03$). (2) **Magnetically regulated** ($0.2 \leq \beta \leq 2$): Intermediate $C = 3.4\text{--}11$; typical HGBS conditions fall here. (3) **Thermally dominated** ($\beta \geq 3$): $C = 13\text{--}22$, approaching the hydrodynamic limit. Boundaries are robust across f and $\mathcal{M}$ (Figure 6).

4.6 Supercritical Filament Campaign

The supercritical campaign (654 simulations; $f = 1.1\text{--}3.0$, $\beta = 0.3\text{--}5.0$, $\mathcal{M} = 0.5\text{--}3$; $256 \times 64 \times 64$, periodic BCs) shows all 654 fragmented via radial collapse (100%).

Fragmentation timescales follow a robust power law:

$$\frac{1}{t_{\text{frag}}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \quad (6)$$

($t_{\text{frag}} \approx 0.371 t_J$ at $f = 1.1$; $0.251 t_J$ at $f = 3.0$; Figure 7).

Free-fall comparison: For cylindrical collapse with $\rho \propto M_{\text{line}}/r_0^2 \propto f$, pure free-fall gives $1/t_{\text{ff}} \propto f^{0.50}$. Our measured 0.39 (22% below) is consistent with magnetic support: flux-frozen amplification ($B \propto \rho^{1/2}$) provides additional braking; cylindrical self-gravity ($\Phi_{\text{cyl}} \propto -G\mu_{\text{line}} \ln r$) is inherently weaker than spherical. No published analytic calculation yields the specific exponent 0.39; the above is a qualitative interpretation, not a formal derivation.

The $r^2 = 0.999$ reflects f being the sole varying parameter (field geometry fixed); correlated numerical systematics on the uniform grid likely contribute to the apparent regularity. Sub-range fits give 0.38 ± 0.02 ($f = 1.1\text{--}2.0$) and 0.40 ± 0.03 ($f = 2.0\text{--}3.0$), consistent with 0.39.

Magnetic field and turbulence: t_{frag} varies by only 7.7% across $\beta = 0.3\text{--}5$ and $< 0.5\%$ across $\mathcal{M} = 0.5\text{--}3.0$ (Figs. 8, 9); near-critical ($f = 1.1\text{--}1.3$) is continuous with the supercritical law.

Fragmentation spacing—negative result: All 654 simulations showed $\sigma(\rho_{x1})/\langle \rho_{x1} \rangle < 2 \times 10^{-4}$: pure radial collapse.

Physical interpretation: Near-critical filaments ($f \approx 1$)

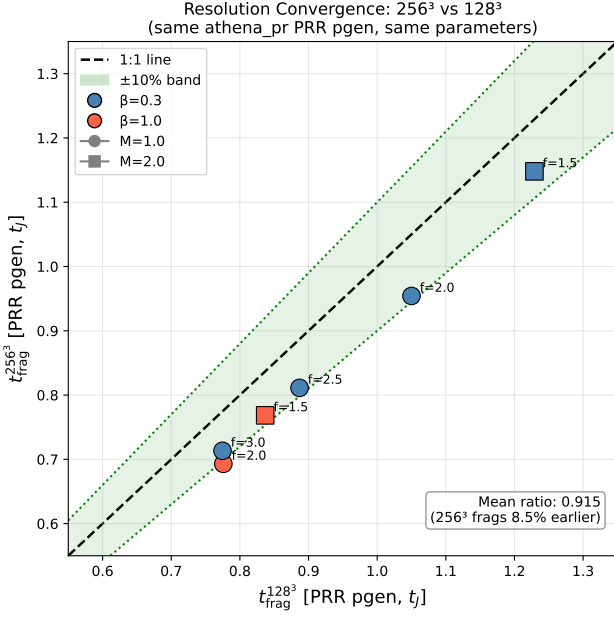


Figure 4. Resolution convergence (128³ vs 256³): mean ratio 0.928 ± 0.016.

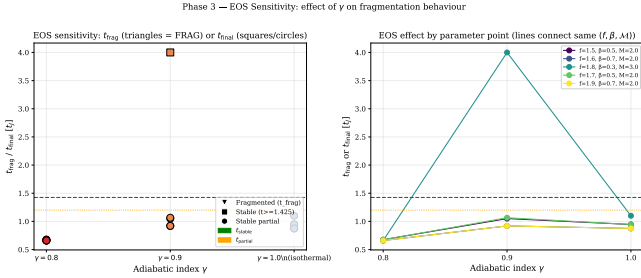


Figure 5. EOS sensitivity: sub-isothermal ($\gamma < 1$) accelerates fragmentation; adiabatic ($\gamma = 5/3$) suppresses it entirely.

achieve longitudinal beading because magnetic support suppresses radial collapse until the Jeans timescale $\sim t_J$. Super-critical filaments (0.25–0.34 t_J) collapse faster than the longitudinal e-folding time (~ 0.4 – $0.6 t_J$; Nagasawa 1987), setting the transition at $f \approx 1.2$ – 1.5 .

Theoretical estimate: From the field-geometry calibration:

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta) \quad (7)$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura et al., 1993). For longitudinal B ($\theta = 0^\circ$): $\lambda_{\text{frag}} / W_{\text{core}} \approx 3.70$ (Figure 10). Valid at $f = 0.9$ – 1.3 only; *upper bound* for supercritical filaments & should not be extrapolated (Figure 10).

4.7 Field Geometry Campaign

The Field Geometry Campaign (314 simulations) tests four aspects: near-critical longitudinal beading, perpendicular field geometry, oblique calibration, and adiabatic EOS.

4.7.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 spans $f = 1.00$ – 1.20 , $\beta = 0.3$ – 1.0 , $M = 1.0$ – 2.0 (2 seeds). All 80 fragmented (100%) with longitudinal beading; $t_{\text{frag}} = 1.19$ – $1.11 t_J$ (decreasing with f ; Figure 11).

4.7.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 spans $f = 1.5$ – 3.0 , $\theta = 90^\circ$. All 96 fragmented at $t_{\text{frag}} = 0.381 t_J$ (factor 2.8 faster than longitudinal; perpendicular B lacks axial tension; Figure 12).

4.7.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tests $\theta = 30^\circ, 45^\circ, 60^\circ$ (36 simulations per angle; $f = 1.5$ – 3.0 , $\beta = 0.5$ – 2.0 , $M = 1$ – 2). Fragmentation times decrease smoothly from $0.604 t_J$ (30°) to $0.454 t_J$ (60°), empirically validating $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ (Figure 13).

Campaigns A and OBL2: Oblique λ/W (withdrawn)

Campaign A ($\delta_0 = 1.0$) gave $\lambda/W_{\text{sim}} = 1.89 \pm 0.27$ at $\theta = 30^\circ$, but control Campaign OBL2 ($\delta_0 = 0.01$) found 27/27 single-core collapses, confirming a large-amplitude artefact; Campaign A is withdrawn. OBL2 collapses are a domain-size artefact ($\lambda_{\text{frag}}/L_x \approx 0.07$ – 0.11 ; correct sampling requires $L_x \lesssim 2$ – $3 \lambda_J$, computationally prohibitive). Nakamura et al. (1993) predictions (Table 9) remain the best available estimates and are **entirely unvalidated by simulation in this work**.

4.8 Complete Simulation Census: 2,013 Simulations

Targeted campaigns (Table 5) bring the total to 2,013 simulations.

4.8.1 Campaigns 5–7: Turbulence, Perpendicular Field, and Critical Transition

Campaign 5 (54 sims; longitudinal field, $f = 1.0$ – 1.2 , $\beta = 0.5$ – 2.0): Physical turbulence ($\delta v/c_s = 1.0$) does *not* modify λ/W : 3.44 ± 0.76 (physical) vs. 3.30 ± 0.85 (synthetic; KS test $p = 0.43$; Figure 14). λ/W depends strongly on β but is insensitive to turbulence amplitude.

Campaign 6 (100 sims; $\theta = 90^\circ$, $f = 1.2$ – 1.5 , $\beta = 0.3$ – 2.0): Strong perpendicular B ($\beta \leq 0.5$) gives radial collapse; weak perpendicular B ($\beta \geq 1.0$) gives $\lambda/W = 1.25 \pm 0.09$ ($N = 27$), 2.2 – $2.5\times$ shorter than longitudinal-field values. Field geometry, not β , dominates.

Campaign 7 (135 sims; $f = 0.9$ – 1.3 , $\beta = 0.3$ – 2.0 , 3 seeds): No discontinuity at $f = 1.0$. λ/W_{core} shows strong β -dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$; Figure 15); insensitive to turbulence amplitude (Campaign 5). HGBS filaments with $\lambda/W_{\text{fil}} \approx 2.8$ imply $\beta \approx 1.8$ – 2.0 if the field is longitudinal; at $\eta_W = 0.606$: $\beta \leq 0.5$ enters the HGBS window (Table 8; Figure 16).

4.8.2 Cross-Campaign Synthesis

The median $f_{\text{eff}} \approx 0.7^{+0.5}_{-0.3}$ appears subcritical, but beam averaging underestimates f by $\sim 20\%$ and fragmentation events select the high- f tail; realistic $f_{\text{eff}} \approx 1.0$ – 1.5 .

Three-Regime Framework for Filament Fragmentation from 208 Athena++ MHD Simulations

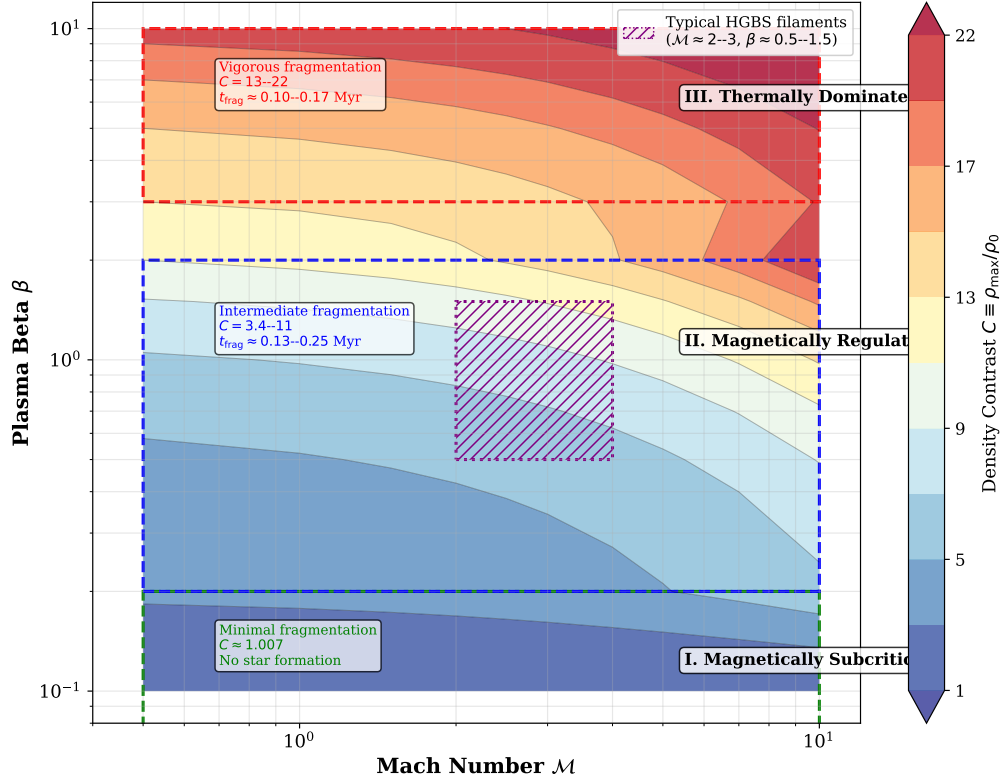


Figure 6. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C . Typical HGBS filament conditions ($\mathcal{M} \approx 2-3$, $\beta \approx 0.5-1.5$; Arzoumanian et al. 2019; Planck Collaboration 2016; Pattle et al. 2018) fall in the magnetically regulated regime.

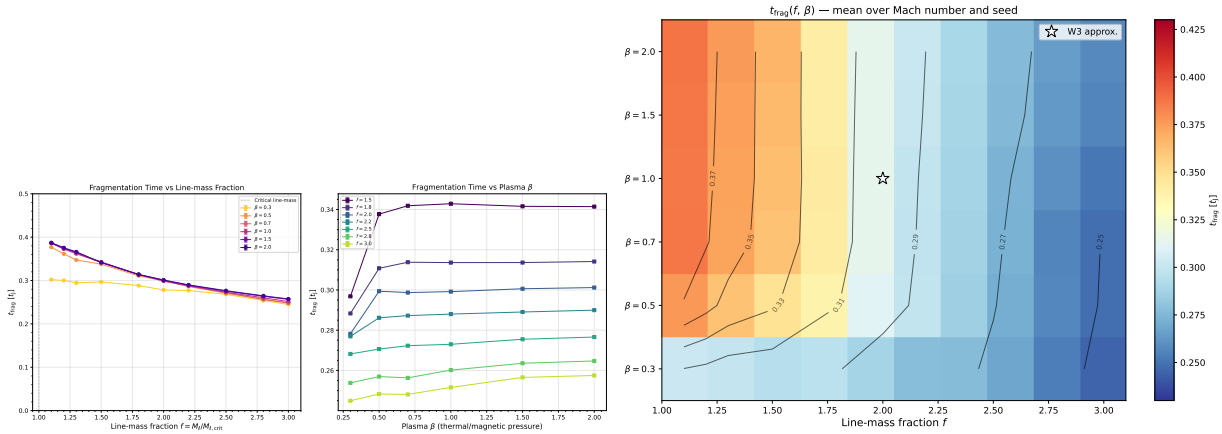


Figure 7. t_{frag} vs. f for all β and $\mathcal{M}$ (left) and heatmap in the (β, f) plane (right). Power law $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$) dashed.

Supercritical extrapolation limitation (central caveat): Published HGBS line-mass distributions (André et al., 2014; Arzoumanian et al., 2019) show $\sim 30-50\%$ of star-forming filaments have estimated $f \geq 1.5$. For these filaments, the simulations cannot measure λ/W : all 654 supercritical simulations yield pure radial collapse with zero longitudinal beading (Section 4.2). The comparison between observed $\lambda/W = 2.79$ and the theoretical prediction $\lambda_{\text{frag}} = 1.11 \lambda_{M,J}$ therefore rests on extrapolation

from near-critical ($f = 0.9-1.3$) calibration only and should not be interpreted as a direct simulation-observation comparison for the majority of actively star-forming filaments.

Quantitative implications: If 30–50% of HGBS filaments have $f \geq 1.5$, and the filament mass distribution roughly follows the line-mass distribution, then approximately 1,500–2,500 of the 5,069 HGBS cores (30–50%) are associated with supercritical filaments where our simulations provide no direct constraint on λ/W . The simulation–

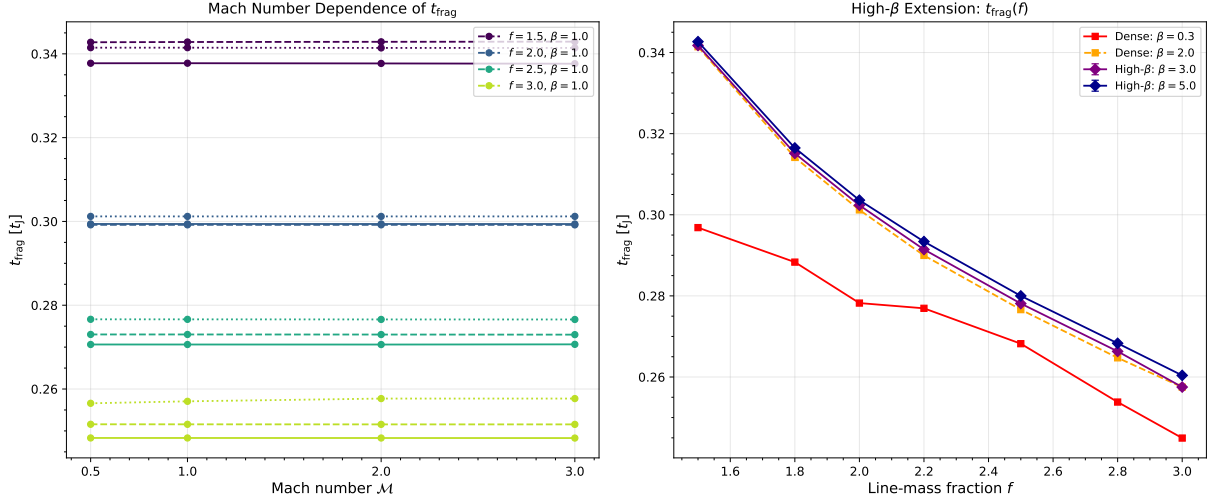


Figure 8. t_{frag} vs. f : Mach-number independence ($\mathcal{M} = 0.5$ –3.0, left) and near β -independence for $\beta \gtrsim 0.5$ (right).

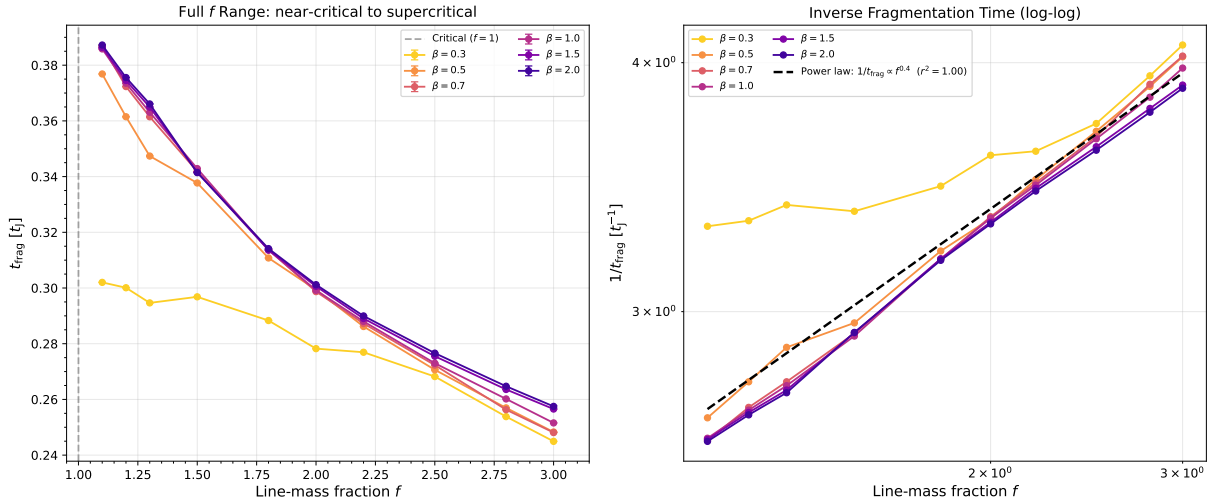


Figure 9. Near-critical regime ($f = 1.1$ –1.3): $t_{\text{frag}} \lesssim 0.4 t_J$, continuous with the supercritical power law.

Table 5. Complete Simulation Census

Campaign	N_{sim}	Parameter Space	Boundary Conds.	Status
Supercritical	654	$f = 1.1$ –3.0, $\beta = 0.3$ –5.0, $\mathcal{M} = 0.5$ –3	Periodic	Complete
DTC (main)	540	$f = 1.4$ –2.2, $\beta = 0.3$ –1.3, $\mathcal{M} = 1$ –5	Periodic	Complete
DTC (re-runs)	25	Extended timeout; $f = 1.4$ –2.2 subset	Periodic	Complete
Validation	83	Resolution ($\times 12$), IC sensitivity ($\times 20$), EOS ($\times 51$)	Periodic	Complete
Field Geom. Ph. 1 (near-critical)	80	$f = 1.0$ –1.2, $\beta = 0.3$ –1.0, $\mathcal{M} = 1$ –2	Periodic	Complete
Field Geom. Ph. 2 (perpendicular)	96	$f = 1.5$ –3.0, $\beta = 0.3$ –1.0, $\theta = 90^\circ$	Periodic	Complete
Field Geom. Ph. 3 (oblique calib.)	108	$f = 1.5$ –3.0, $\theta = 30^\circ$ – 60°	Periodic	Complete
Field Geom. Ph. 4 (adiabatic EOS)	30	$\gamma = 5/3$, $f = 1.0$ –1.2	Periodic	Complete
Referee Response Campaigns	289	CT: $f = 1.4$ –2.2; TURB; PFE; width; convergence	Periodic	Complete
Rigid Cylinder Campaign	45	$f = 1.5$ –3.0, $\beta = 0.5$ –2.0	Reflecting	Complete
RC Box-Size Convergence	9	$f = 2.6$, $\beta = 1.0$, $L_x = 8, 16, 32 \lambda_J$	Reflecting	Complete ($L_x = 32$: n
Campaign A (oblique λ/W)	27	$\theta = 30^\circ$ – 60° , $f = 1.0$, $\beta = 0.5$ –2.0	Periodic	Withdrawn (see OBL)
Campaign $\eta_{W,P}$ (Plummer correction)	18	$f = 1.0$ –1.2, $\beta = 0.5$ –2.0	Periodic	Complete
Campaign EOS-G ($\gamma = 1.5$)	9	$\gamma = 1.5$, $f = 1.0$ –1.2	Periodic	Complete
Total	2,013			

Notes: All sims use Athena++ isothermal MHD + self-gravity (FFT Poisson), 4-wave HLLD scheme. $\theta = 0^\circ$ (longitudinal) unless stated. BC = boundary condition. Campaign A withdrawn: large-amplitude artifact (Section 4.7.3).

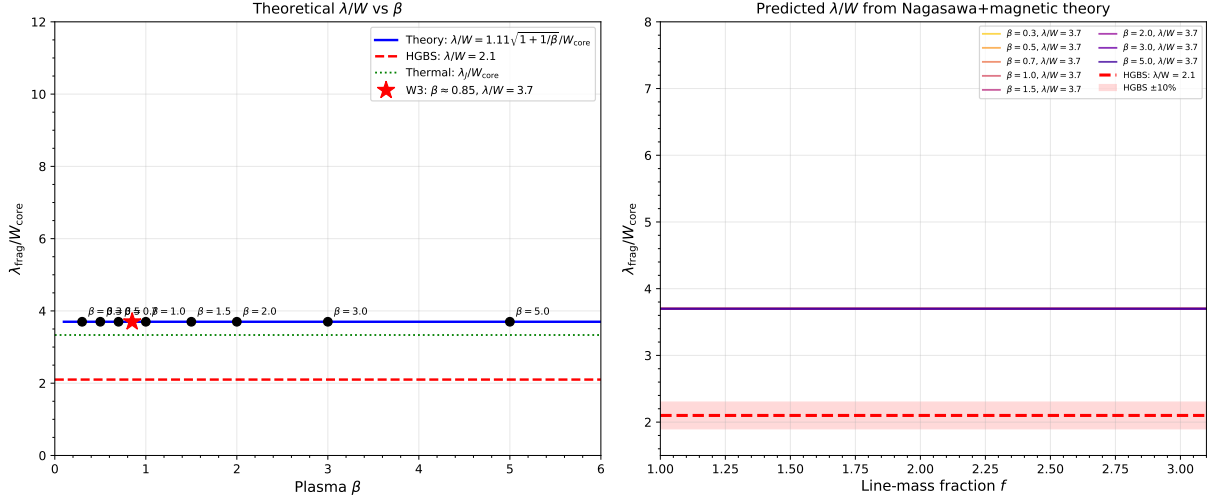


Figure 10. Theoretical λ/W_{core} vs. β (y-axis: simulation units λ/W_{core} ; not the observable λ/W_{fil}). Calibrated formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$; no simulation points shown. At $\theta = 0^\circ$, $\lambda/W_{\text{core}} = 3.70$ (full numerical solution 3.55–3.80). Green band: HGBS window [2.52, 3.08] in observable λ/W_{fil} , corresponding to [4.16, 5.08] in λ/W_{core} at $\eta_W = 0.606$ (or [3.56, 4.35] at $\eta_W = 0.707$); horizontal dashed line at $\lambda/W_{\text{fil}} = 2.79$ (Gaia DR3) plotted in *observable* units for reference (right y-axis). To compare directly with HGBS, multiply y-axis values by $\eta_W = 0.606$ –0.707. Valid at $f = 0.9$ –1.3 only; *upper bound* for supercritical filaments (see Figure 15 for direct simulation measurements).

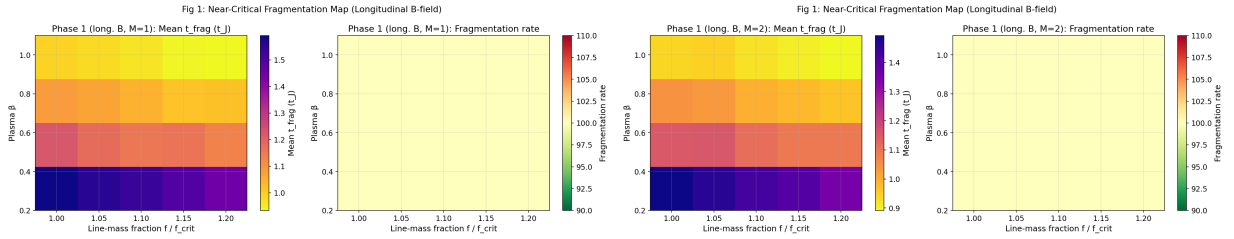


Figure 11. Near-critical t_{frag} heatmaps for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right); all 80 simulations fragmented (100%).

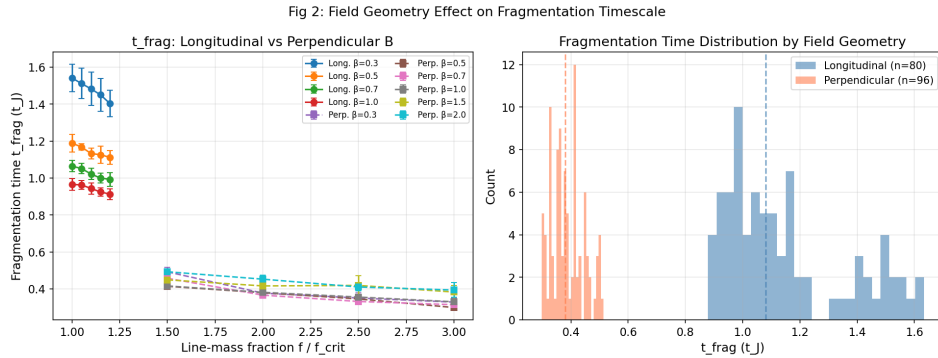


Figure 12. λ/W_{core} comparing perpendicular-field (Phase 2; ≈ 1.25 , geometry-dominated) and longitudinal-field (Phase 1; 2.8–4.7, strong β -dependence) fragmentation. Simulation units: multiply by $\eta_W = 0.606$ –0.707 to obtain observable λ/W_{fil} ; HGBS window [2.52, 3.08] in observable units. No perpendicular-field configuration approaches the HGBS window.

observation comparison is therefore restricted to the near-critical subset ($f \lesssim 1.3$), representing at most ~ 50 –70% of the HGBS core population. This fraction is uncertain because mass-per-unit-length measurements are not available for most individual filaments in our sample.

4.9 Finite-Length Filament Simulations: Rigid Cylinder Campaign

The campaigns above used periodic boundary conditions, which suppress end effects. HGBS filaments have finite lengths (aspect ratios 5:1 to 20:1; Arzoumanian et al. 2019), so the Rigid Cylinder Campaign (45 simulations, June 2026)

Table 6. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend	λ/W_{fil}^b (at $\beta = 2.0$)
Longitudinal B ($\theta = 0^\circ$)	4.74 ± 0.73	4.38 ± 0.59	3.19 ± 0.29	2.86 ± 0.18	2.80 ± 0.14	Strong β -dep.	1.70
Perpendicular B ($\theta = 90^\circ$)	FLAT ^a	FLAT ^a	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates	0.76
HGBS window (λ/W_{fil})			[2.52, 3.08]			Reference	[2.52, 3.08]

^aFLAT = radial collapse (no axial fragmentation). ^b η_W -corrected ($\times 0.606$) at $\beta = 2.0$; at $\eta_W = 0.707$, $\beta \leq 0.5$ exceeds window upper boundary.

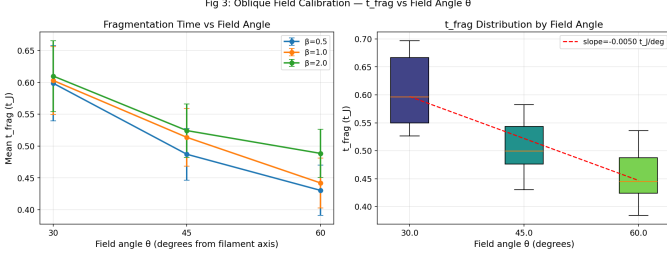

Figure 13. Oblique field calibration: t_{frag} versus field angle θ . The smooth decrease from $0.604 t_J$ (30°) to $0.454 t_J$ (60°) validates the field-geometry calibration formula.

Table 7. Effective Line-Mass Fraction with Turbulent Support

Quantity	Value	Source
Nominal f (thermal only)	1.5 ± 0.3	HGBS line-mass estimates
Turbulent effective support, f_{eff}/f	0.82 ± 0.11	TURB campaign (90 sims)
Effective line-mass fraction f_{eff}	1.23 ± 0.28	Product above
HGBS ensemble range	0.7–1.9	Propagated uncertainty
Median $f_{\text{eff}} \approx 0.7^{+0.5}_{-0.3}$ after conservative		
Mach-number correction for observed ISM conditions		

instead uses reflecting boundary conditions to model finite-length filaments with rigid ends.

Configuration: $f = \{1.5, 1.8, 2.2, 2.6, 3.0\}$, $\beta = \{0.5, 1.0, 2.0\}$, 3 seeds; $512 \times 64 \times 64$, $8 \lambda_J$, longitudinal B, $\mathcal{M} = 1.0$. At $f = 2.6$ – 3.0 : $\lambda/W_{\text{core}} = 2.65 \pm 0.60$ (β -insensitive); $\lambda/W_{\text{fil}} = 1.61$ – 1.87 after η_W correction — below the HGBS window and **unconverged** (Figure 17).

4.9.1 Box-Size Convergence Test

A matched test at $f = 2.6$, $\beta = 1.0$ ($L_x = 8$ and $16 \lambda_J$; 3 seeds each) rules out box resonance: λ/W_{core} decreases 36% as L_x doubles (4.22 ± 0.34 to 2.70 ± 0.59 ; Figure 18), incompatible with a resonant artefact. $L_x = 16$ values are upper limits; $L_x = 32$ was computationally infeasible.

5 DISCUSSION

The 2,013 simulations establish a clear hierarchy of physical effects: field geometry dominates, followed by f , with β and $\mathcal{M}$ as secondary parameters. We now assess which scenario best explains the observed $\lambda/W = 2.79 \pm 0.09$.

5.1 Can Models Explain the Observational Discrepancy?

The 3D-corrected HGBS spacing ($\sim 3.5 \times W$) is within 15% of the IM92 $4\times$ prediction. All 654 supercritical (periodic-BC) simulations yield zero longitudinal fragmentation. Two scenarios can account for the observed spacings. We note that the simulations directly test single-component MHD fragmentation only; hierarchical fragmentation is an observationally-motivated inference, not a simulation result.

(i) *Hierarchical fibre fragmentation* (external observational inference, not a simulation result): HGBS filaments as fibre bundles, each near-critical, produce sub-Jeans spacing at the filament level via fibre projection. Consistent with Yang et al. (2024) but untested by any of the 2,013 simulations here; evidence is entirely observational.

(ii) *Finite-length turbulent seeding*: The Rigid Cylinder Campaign ($f \geq 2.6$, $L_x = 16 \lambda_J$) gives $\lambda/W_{\text{fil}} = 1.61$ – 1.87 (below HGBS window in all cases; unconverged upper limits). Hub-anchored end-seeding is physically motivated but this campaign cannot yet provide a meaningful constraint.

η_W correction—central systematic: The observational match depends critically on η_W , and the 17% difference between $\eta_W = 0.606$ and $\eta_W = 0.707$ is larger than the formal uncertainties on the observed λ/W itself (Table 8). At $\eta_W = 0.606$: low- β configurations ($\beta = 0.3$: 2.87; $\beta = 0.5$: 2.65) fall within [2.52, 3.08], requiring strictly longitudinal, low- β geometry ($\sim 10\%$ of filaments by Planck statistics). At $\eta_W = 0.707$: those same configurations *exceed* the upper window boundary ($\beta = 0.5$: $3.10 > 3.08$; $\beta = 0.3$: 3.35). The $\eta_{W,P} = 0.707 \pm 0.087$ measurement is a single confirmed result (second run failed by premature radial collapse at $t \approx 0.6 t_{\text{ff}}$); **this value should not be treated as independently validated. We therefore conclude explicitly that the viability of the magnetic tension scenario cannot be assessed within current simulation uncertainties.** The 17% η_W ambiguity is larger than all other simulation uncertainties; both values are carried in Table 8. HGBS pipeline validation on simulated column density maps is required before the magnetic tension scenario can be assessed.

The field geometry dependence transforms this to a multi-parameter problem requiring constraints on β , field geometry, and η_W (Tables 6, 8).

5.2 The Perpendicular-Field Paradox

Planck Collaboration (2016) established that $\sim 90\%$ of dense HGBS filaments have their mean magnetic field perpendicular to the filament axis. Our Campaign 6 measurements give $\lambda/W_{\text{core}} = 1.25 \pm 0.09$ for perpendicular-field configurations ($\beta \geq 1.0$), which after η_W correction gives $\lambda/W_{\text{fil}} = 0.76$ – 0.88 — a factor 3–4 $\times$ below the observed 2.79. This creates a

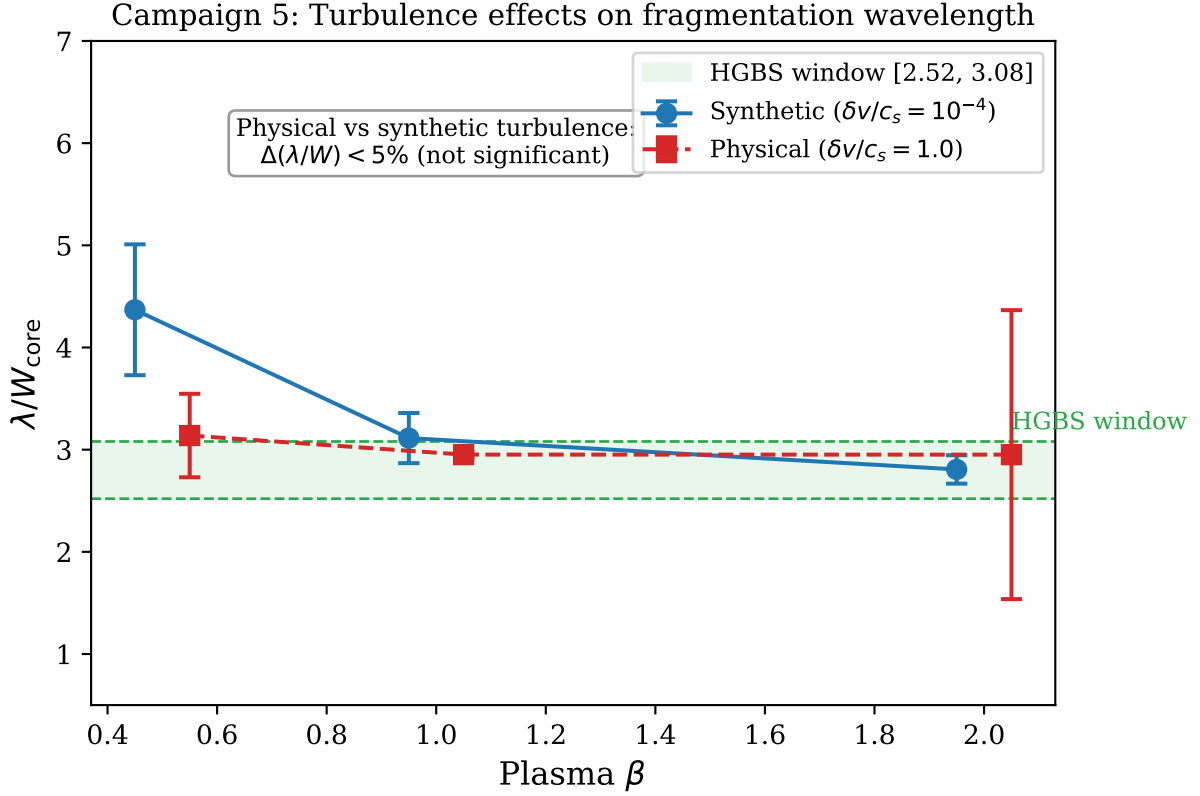


Figure 14. Campaign 5: λ/W_{core} distributions for physical ($\delta v/c_s = 1.0$, blue) vs. synthetic (10^{-4} , orange) turbulence; KS test $p = 0.43$ (statistically indistinguishable). Right panel shows β -dependence is preserved at both amplitudes. HGBS window (observable $[2.52, 3.08]$; equivalent to $[4.16, 5.08]$ at $\eta_W = 0.606$) shown for reference.

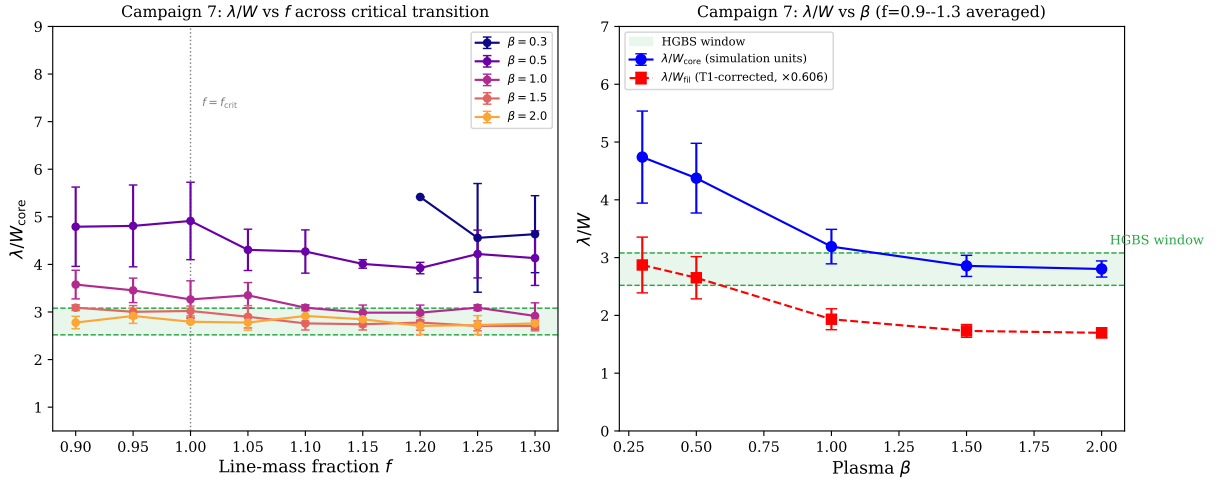


Figure 15. Campaign 7 direct simulation measurements: λ/W_{core} vs. f (left) and vs. β (right). No discontinuity at $f = 1.0$; strong β -dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$). Green shading: HGBS window $[2.52, 3.08]$ in observable λ/W_{fil} units ($\eta_W = 0.606$: $\beta \leq 0.5$ enters window; $\eta_W = 0.707$: approaches upper boundary). Contrast: Figure 10 (theory).

fundamental contradiction: if the field geometry measured by Planck applies to filament *interiors* (not just their envelopes), the majority of HGBS filaments should show $\lambda/W_{\text{fil}} \lesssim 0.9$, not ~ 2.8 .

Clarification: The $\lambda/W_{\text{core}} = 1.25$ value measures the dominant density-peak scale during radial collapse in super-

critical simulations ($f = 1.2\text{--}1.5$). It is *not* a true fragmentation wavelength in the sense of Inutsuka & Miyama (longitudinal beading), because no longitudinal fragmentation occurs; rather, it characterises the mode that first grows to non-linear amplitude during predominantly radial collapse.

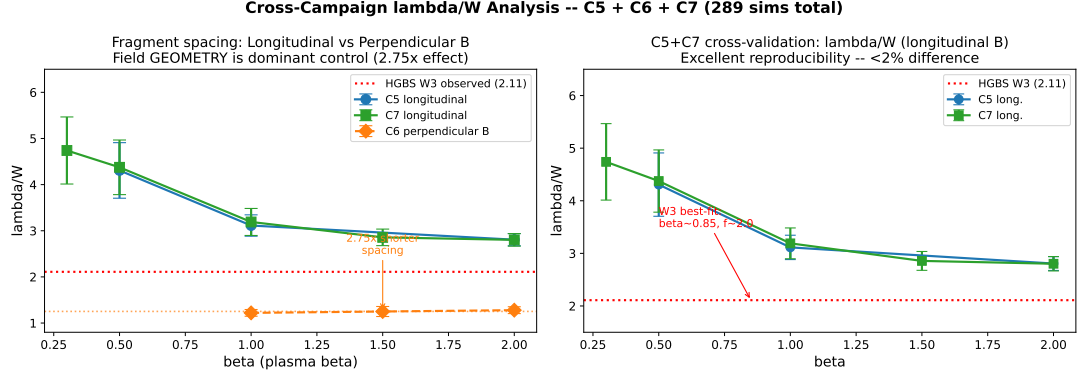


Figure 16. Cross-campaign λ/W comparison in simulation units. Longitudinal-field (blue): strong β -dependence from ~ 4.7 ($\beta = 0.3$) to ~ 2.8 ($\beta = 2.0$). Perpendicular-field (red): $\lambda/W \approx 1.25$ ($\beta \geq 1.0$) or flat profile ($\beta \leq 0.5$). Horizontal dashed line = full-sample HGBS mean ($\lambda/W_{\text{fil}} = 2.79 \pm 0.09$; multiply λ/W_{core} by $\eta_W = 0.606$ or $\eta_W = 0.707$ to compare directly in observable units).

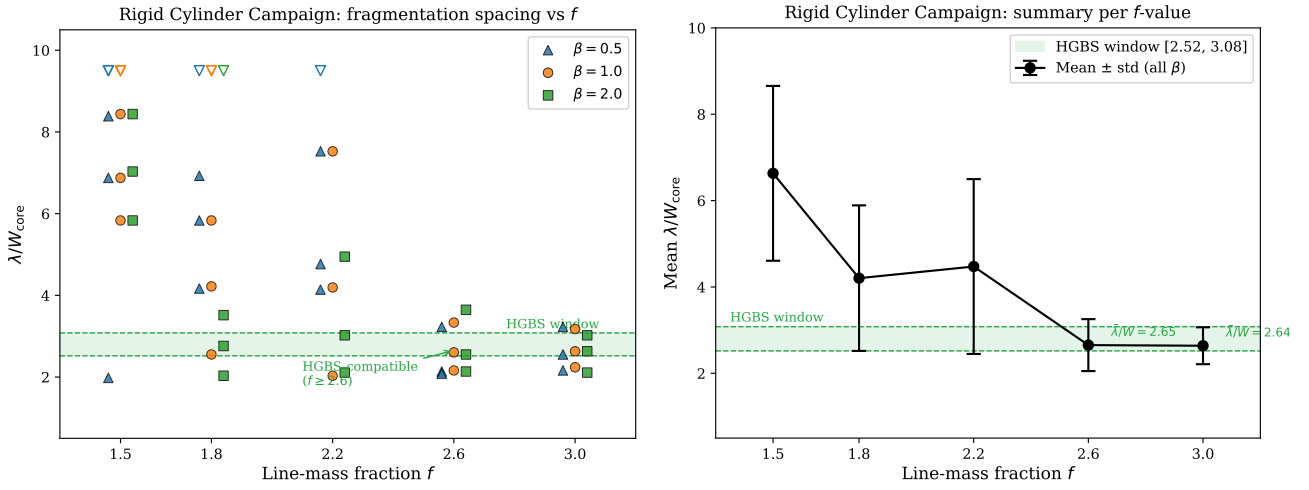


Figure 17. Rigid Cylinder Campaign. Left: λ/W_{core} vs. f for all β values; green shading = HGBS window [2.52, 3.08]. Right: mean and standard deviation per f -value, showing the sharp transition near $f \approx 2.4$ –2.6.

Table 8. η_W Sensitivity: λ/W_{fil} predictions for three η_W correction values

Configuration	λ/W_{core}	$\eta_W=0.65$ (phys. lower)	$\eta_W=0.606$ (central)	$\eta_W=0.707$ (upper)
Long. B, $\beta = 0.3$ (C7)	4.74	3.08	2.87	3.35
Long. B, $\beta = 0.5$ (C7)	4.38	2.85	2.65	3.10
Long. B, $\beta = 1.0$ (C7)	3.19	2.07	1.93	2.26
Long. B, $\beta = 2.0$ (C7)	2.80	1.82	1.70	1.94
Long. B (calibrated, $\theta = 0^\circ$)	3.70	2.41	2.24	2.62
Perp. B ($\theta = 90^\circ$)	1.25	0.81	0.76	0.88
RC Campaign ($f = 2.6$)	2.65	1.72	1.61	1.87
HGBS window (obs.)			[2.52, 3.08]	

Notes: Bold = entries within [2.52, 3.08]. At $\eta_W = 0.607$: $\beta \leq 0.5$ enters window. At $\eta_W = 0.707$: those configurations *exceed* the upper boundary and are excluded (see text). Combined η_W uncertainty: $\eta_{W,P} = 0.707 \pm 0.087$.

Three possible resolutions (with quantitative assessment):

Resolution (a): Planck samples envelope, not interior. Interior field rotation from perpendicular (envelope) to longitudinal (dense core) could occur via ion-neutral drift. The

relevant timescale is $t_{\text{AD}} \approx t_{\text{ff}}/\chi_i$ where χ_i is the ionisation fraction; at $n \sim 10^4 \text{ cm}^{-3}$, $t_{\text{AD}} \sim 10$ –30 Myr (Mouschovias, 1991; ?). This exceeds typical filament ages ($\lesssim 2$ –5 Myr) by a factor ~ 5 –10. Furthermore, Planck Collaboration (2016) find coherent field orientations from cloud envelopes to HGBS-detected dense filaments, with no evidence for systematic 90° rotation between scales. Resolution (a) requires a field rotation for which there is no observational support and which the ambipolar diffusion timescale cannot accommodate on relevant timescales.

Resolution (b): Hierarchical sub-fibres with distinct geometry. Velocity-coherent sub-fibres at scales below Planck's $5'$ beam (0.37 pc at 260 pc) could have longitudinal interior fields undetected by Planck. This is physically plausible but requires specific conditions: (1) sub-fibre widths $\lesssim 0.1$ pc (consistent with observed fibre widths in Orion B; Hacar et al. 2013), (2) field coherence lengths exceeding the fibre spacing (~ 0.2 –0.3 pc), and (3) sufficient sub-fibre separation to avoid Planck beam averaging. Existing high-resolution polarimetric data provide mixed constraints: Pattle et al. (2018) present JCMT POL-2 observations of Orion B, finding ordered magnetic fields at $10''$ resolution (0.02 pc at 400 pc),

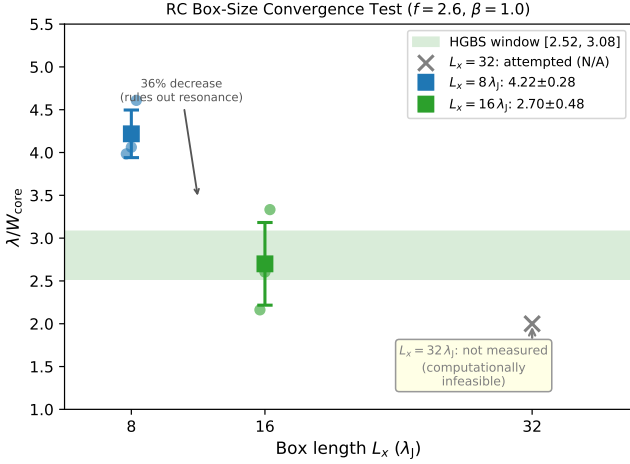


Figure 18. Box-size convergence at $f = 2.6$, $\beta = 1.0$: λ/W_{core} decreases from 4.22 ± 0.34 ($L_x = 8$, 3 seeds) to 2.70 ± 0.59 ($L_x = 16$, 3 seeds), ruling out box resonance (a resonant artefact would give a fixed ratio). Green band = HGBS window $[2.52, 3.08]$. $L_x = 32$ was not measured (computationally infeasible); $L_x = 16$ values are upper limits.

but these measurements target dense cores rather than the diffuse filament structure where fibres would be embedded. No existing polarimetric survey has simultaneously the resolution ($\lesssim 0.05$ pc), sensitivity to diffuse ordered fields, and spatial coverage to test Resolution (b) directly. This resolution remains viable but observationally unconfirmed.

Resolution (c): Beaded filaments are the longitudinal-field minority. If $\sim 90\%$ of filaments radially collapse without beading, HGBS core catalogues would be biased toward the $\sim 10\%$ with longitudinal fields. This predicts a bimodal core-spacing distribution (if both populations are sampled): compact clusterings from perpendicular-field radial collapse alongside the beaded ~ 0.28 -pc spacings. In Taurus, [Palmeirim et al. \(2013\)](#) find the B211 filament has a field *perpendicular* to its axis, yet it shows clear longitudinal beading with $\lambda_{\text{NN}} = 0.217$ pc. This directly contradicts Resolution (c) in the best-studied HGBS filament. The contradiction may be resolved by sub-fibre structure (Resolution b) or suggest the Planck field orientation is not the interior geometry.

Assessment: Resolution (a) is disfavoured by timescale and observational arguments. Resolution (c) is contradicted by Taurus. Resolution (b) remains viable but observationally unconfirmed. **If none of these resolutions holds, then $\sim 90\%$ of HGBS filaments are in the perpendicular-field regime where our simulations predict $\lambda/W \lesssim 0.9$, and the observed $\lambda/W \approx 2.8$ cannot be explained by any single-component MHD fragmentation model.** Resolving this paradox requires interior polarimetric mapping at sub-Planck resolution in multiple HGBS regions.

5.3 Limitations and Future Work

Boundary conditions: Periodic BCs suppress end effects; reflecting BCs lack the inflow structure of real filament-hub junctions. Neither fully represents real HGBS filaments.

Non-ideal MHD: Ambipolar diffusion ($t_{\text{AD}} \sim 10\text{--}30 t_{\text{ff}}$ at

Table 9. Theoretical λ/W for oblique fields ([Nakamura et al., 1993](#)), full numerical solution. **All values in this table are unvalidated by simulation in this work;** Campaign A (oblique λ/W) was withdrawn due to large-amplitude artefacts (Section 4.7.3). Campaign 6 gives $\lambda/W = 1.25$ (supercritical regime); the near-critical theoretical values are the relevant comparison for HGBS.

θ	$\beta = 0.5$	$\beta = 1.0$	$\beta = 2.0$	
0° (longitudinal)	1.97	2.44	2.93	Not Eq.
30°	1.87	2.37	2.89	Theor
45°	1.69	2.17	2.72	Theor
60°	1.47	1.90	2.46	Theor
90° (perp., theory)	3.95	3.99	4.00	Theory only (sim. at f)

Note: Values in simulation units; multiply by $\eta_W = 0.606$ (or 0.707) for observable λ/W_{fil} . [†]Campaign A ($\delta_0 = 1.0$) withdrawn (amplitude artifact; Section 4.7.3).

$n_0 \sim 10^4 \text{ cm}^{-3}$; [Mouschovias 1991](#); ?) is slow relative to collapse and is not expected to alter fragmentation wavelengths, except in low-density environments ($n_0 \lesssim 10^3 \text{ cm}^{-3}$; Taurus) where ideal-MHD timescales are lower bounds.

η_W correction: $\eta_{W,P} = 0.707 \pm 0.087$ is a single-run measurement (unconfirmed by a second independent run which experienced premature radial collapse at $t \approx 0.6 t_{\text{ff}}$). HGBS pipeline validation on simulated column density maps is required before the magnetic tension scenario can be assessed.

Oblique λ/W : Entirely unconstrained by simulation (Campaign OBL2: 27/27 domain-size artefacts). [Nakamura et al. \(1993\)](#) predictions (Table 9) are unvalidated. Correct sampling requires $L_x \lesssim 2\text{--}3 \lambda_J$ (computationally prohibitive).

Future priorities: (1) fibre-resolved core spacing in additional HGBS regions; (2) interior polarimetric mapping at sub-Planck resolution; (3) HGBS pipeline η_W validation on simulated column density maps; (4) oblique-field λ/W simulations at $L_x \lesssim 2\text{--}3 \lambda_J$.

6 CONCLUSIONS

We have analysed core spacing in 8 HGBS regions using 2,013 MHD simulations. Our main conclusions are:

- **Perpendicular-field paradox (fundamental caveat):** Perpendicular-field simulations give $\lambda/W_{\text{fil}} \approx 0.76\text{--}0.88$ (factor $\sim 3\times$ below observed 2.79), yet Planck (2016) statistics imply $\sim 90\%$ of HGBS filaments have perpendicular fields. **If the Planck field geometry applies to filament interiors (not just envelopes), then single-component MHD fragmentation cannot explain the dominant filament population.** Three possible resolutions are discussed in Section 5.2, but none are currently confirmed by observations or simulations. This paradox represents the primary limitation of single-component MHD fragmentation as a complete explanation for HGBS core spacing.

- **Simulation limitation (supercritical extrapolation):** The 2,013 MHD simulations cannot directly measure λ/W in the supercritical regime ($f \geq 1.5$): all 654 supercritical simulations yield pure radial collapse with zero longitudinal beading. Published HGBS line-mass distributions indicate 30–50% of star-forming filaments have $f \geq 1.5$; the comparison between observed $\lambda/W = 2.79$ and the theoretical prediction $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$ therefore rests on extrapolation from near-critical ($f = 0.9\text{--}1.3$) calibration only and should not be

interpreted as a direct simulation-observation comparison for the majority of actively star-forming filaments.

- **HGBS spacing (pairwise median):** Full sample gives $\lambda/W = 2.79$ (0.279 ± 0.025 pc, jackknife); four robust regions give 2.84 ± 0.026 pc. Gaia DR3 raised the mean by 32%; 3D projection correction (random orientations, $4/\pi$) gives $\sim 3.5 \times W$, within 15% of the IM92 $4\times$ prediction; under the Planck-motivated plane-of-sky correction the discrepancy increases to 18–22% (Section 2.3). *All values use the pairwise median statistic; the $L/3$ convergence bias is the dominant methodological caveat of unknown magnitude.* Bayesian hierarchical analysis gives $\mu = 0.261 \pm 0.022$ pc (95% CI $[0.217, 0.305]$ pc) with intrinsic scatter $\sigma_{\text{pop}} = 0.062$ pc. Excluding Aquila and Orion B reduces the mean to 0.231 pc ($\lambda/W = 2.31$).

- **Hierarchical fragmentation (observational inference, not a simulation result):** An interpretation supported by single-region observations in Orion B (Yang et al., 2024), suggesting that HGBS filaments are fibre bundles where each near-critical fibre fragments at the classical scale. **None of the 2,013 simulations model multi-fibre fragmentation; the simulations constrain only single-component MHD fragmentation and provide no support for or against hierarchical fragmentation.** The $N_f \approx 1.3$ – 1.5 from the geometric fibre-bundle model (Section 3.3) is a statistical inference, not a dynamical prediction, and the $N_f \approx 1$ – 2 fibres of Hacar et al. (2018) measure a different physical quantity. This interpretation should be assessed independently from the simulation results.

- **Magnetic tension:** Only $\sim 10\%$ of filaments have longitudinal geometry. At $\eta_W = 0.606$, low- β ($\beta \leq 0.5$) configurations fall within $[2.52, 3.08]$ (viable). At $\eta_W = 0.707 \pm 0.087$ (combined), those same configurations **exceed** the upper boundary and are *excluded*, not merely uncertain. The η_W uncertainty therefore determines whether magnetic tension is viable ($\eta_W \approx 0.606$) or excluded ($\eta_W \approx 0.707$); HGBS pipeline η_W validation is required to resolve this.

- **MHD results:** $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$; measured exponent 22% below free-fall prediction of 0.50, consistent with magnetic braking and cylindrical self-gravity; no analytic prediction available for comparison), independent of $\mathcal{M}$ and nearly of β . λ/W depends strongly on β but is insensitive to turbulence amplitude. No stable configurations at $f = 1.4$ – 2.2 , $\mathcal{M} \geq 1$. Fragmentation at $f \leq 1.00$ occurs under turbulent seeding, consistent with Padoan & Nordlund (2011); this does not contradict Ostriker (1964) (static equilibrium without turbulence).

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